SE (LAB-05)

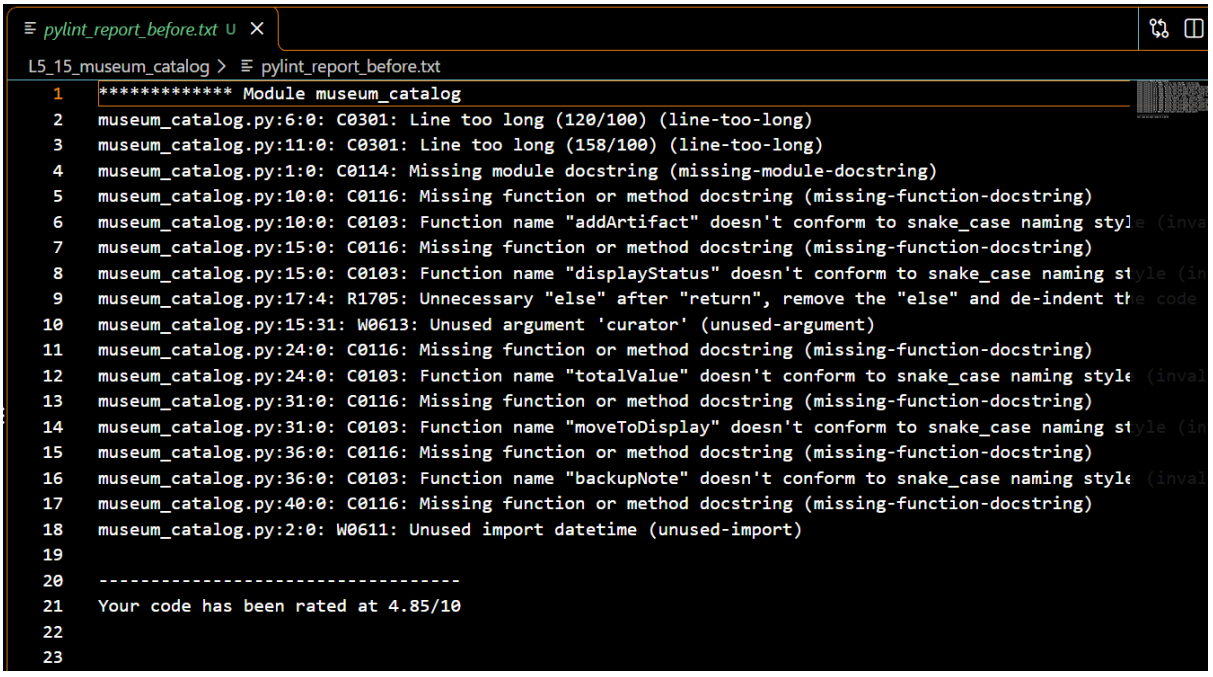
Static Code Analysis

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SRN: PES2UG24CS452
SECTION: H

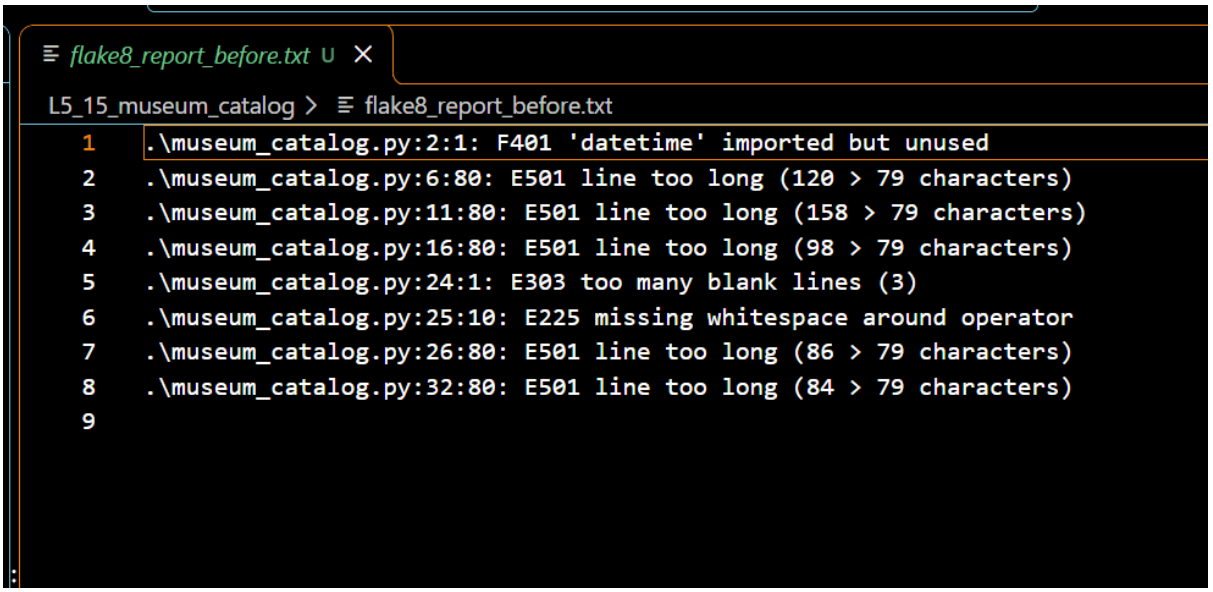
GITHUB: https://github.com/AbdullahMahdi29/L5_15_museum_catalog.git

Before report:

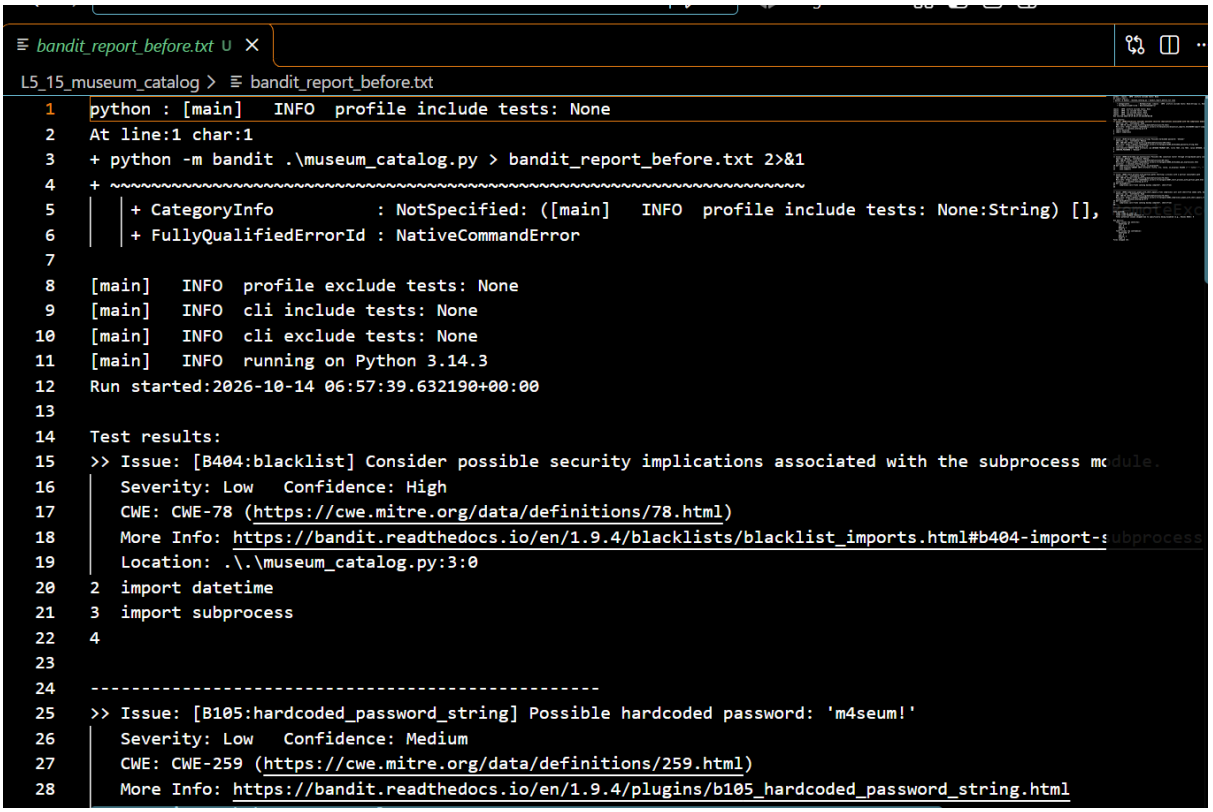
1. Pylint:



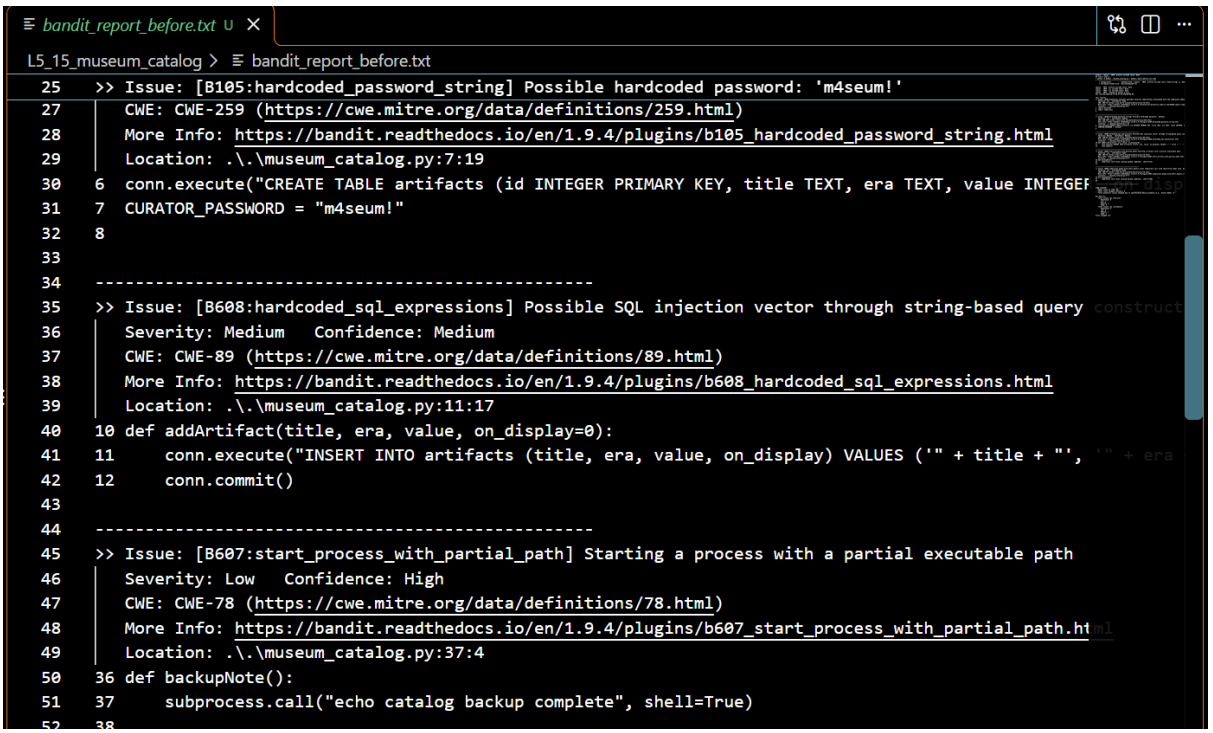
2. Flake 8:


flake8_report_before.txt U
L5_15_museum_catalog > flake8_report_before.txt
1 .\museum_catalog.py:2:1: F401 'datetime' imported but unused
2 .\museum_catalog.py:6:80: E501 line too long (120 > 79 characters)
3 .\museum_catalog.py:11:80: E501 line too long (158 > 79 characters)
4 .\museum_catalog.py:16:80: E501 line too long (98 > 79 characters)
5 .\museum_catalog.py:24:1: E303 too many blank lines (3)
6 .\museum_catalog.py:25:10: E225 missing whitespace around operator
7 .\museum_catalog.py:26:80: E501 line too long (86 > 79 characters)
8 .\museum_catalog.py:32:80: E501 line too long (84 > 79 characters)
9

3. Bandit:



```
bandit_report_before.txt U
L5_15_museum_catalog > bandit_report_before.txt
1  python : [main]   INFO  profile include tests: None
2  At line:1 char:1
3  + python -m bandit .\museum_catalog.py > bandit_report_before.txt 2>&1
4  + ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
5      + CategoryInfo          : NotSpecified: ([main]   INFO  profile include tests: None:String) [],
6      + FullyQualifiedErrorId : NativeCommandError
7
8  [main]   INFO  profile exclude tests: None
9  [main]   INFO  cli include tests: None
10 [main]   INFO  cli exclude tests: None
11 [main]   INFO  running on Python 3.14.3
12 Run started:2026-10-14 06:57:39.632190+00:00
13
14 Test results:
15 >> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
16    Severity: Low   Confidence: High
17    CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
18    More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
19    Location: .\.\museum_catalog.py:3:0
20 2  import datetime
21 3  import subprocess
22 4
23
24 --------------------------------------------------
25 >> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'm4seum!'
26    Severity: Low   Confidence: Medium
27    CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
28    More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
```



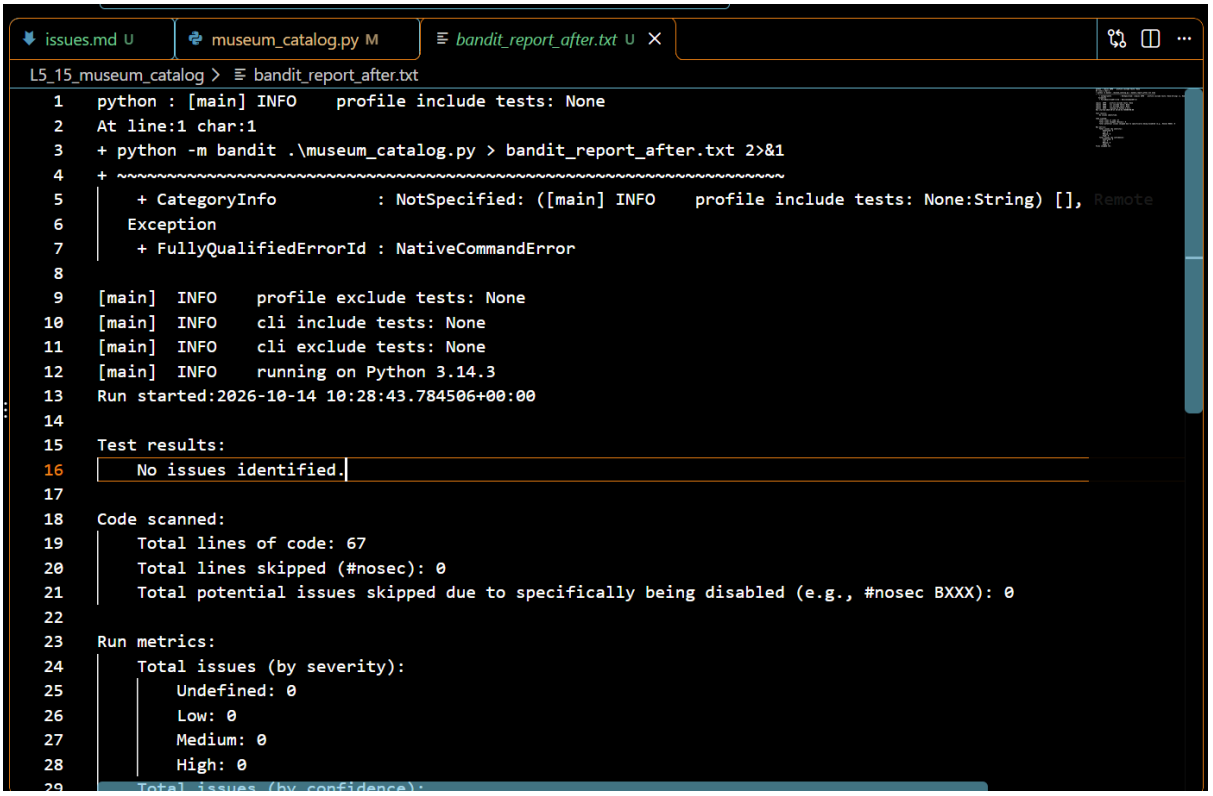
```
bandit_report_before.txt U
L5_15_museum_catalog > bandit_report_before.txt
25 >> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'm4seum!'
27    CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
28    More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
29    Location: .\.\museum_catalog.py:7:19
30 6  conn.execute("CREATE TABLE artifacts (id INTEGER PRIMARY KEY, title TEXT, era TEXT, value INTEGER
31 7  CURATOR_PASSWORD = "m4seum!"
32 8
33
34 --------------------------------------------------
35 >> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construct
36    Severity: Medium   Confidence: Medium
37    CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
38    More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
39    Location: .\.\museum_catalog.py:11:17
40 10 def addArtifact(title, era, value, on_display=0):
41 11     conn.execute("INSERT INTO artifacts (title, era, value, on_display) VALUES ('" + title + "', '" + era
42 12     conn.commit()
43
44 --------------------------------------------------
45 >> Issue: [B607:start_process_with_partial_path] Starting a process with a partial executable path
46    Severity: Low   Confidence: High
47    CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
48    More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b607_start_process_with_partial_path.html
49    Location: .\.\museum_catalog.py:37:4
50 36 def backupNote():
51 37     subprocess.call("echo catalog backup complete", shell=True)
52 38
```

After report:

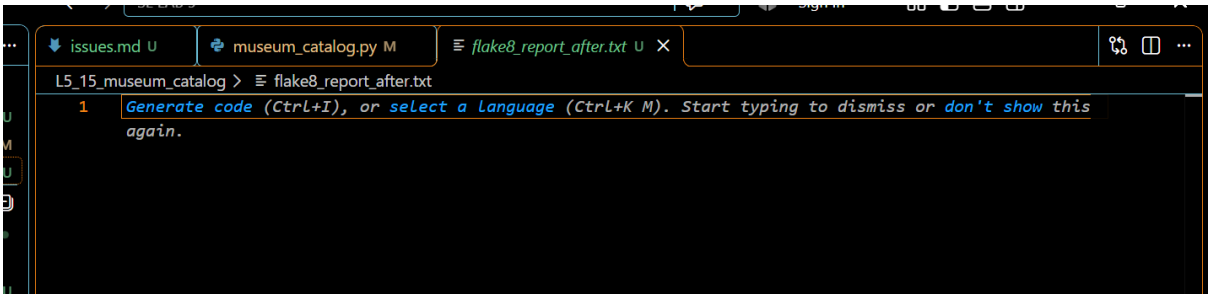
1. Pylint:



2. Bandit:



3. Flake 8:


issues.md U
museum_catalog.py M
flake8_report_after.txt U
L5_15_museum_catalog > flake8_report_after.txt
1
Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing to dismiss or don't show this again.